EXPERIMENT NO: 6

TITLE: PID Controller Implementation Tuning and response study of First-Order & Second-Order systems.

AIM: To implement a PID controller in MATLAB/Simulink, tune its parameters, and study the closed-loop response of first-order and second-order systems for various inputs. The experiment focuses on analyzing the effects of proportional, integral, and derivative gains on system performance, including rise time, settling time, overshoot, and steady-state error.

APPARATUS:

Sr.No.	Equipment	Specification	Version
1	MATLAB IDE	Matlab Simulink model for circuit analysis	R2024a, R2025a

THEORY: PID Controller Implementation, Tuning, and Response Study

1. Introduction

A PID controller (Proportional-Integral-Derivative) is a widely used feedback control mechanism in industrial and academic applications. It continuously calculates an error signal, which is the difference between a desired set point and the measured process variable, and applies corrective control to minimize this error.

The PID control law is given by:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt}$$

Where:

- $u(t)$ = controller output
- $e(t)$ = error = setpoint − process variable
- K_p = proportional gain
- K_i = integral gain
- K_d = derivative gain

2. First-Order System

A first-order system has a single energy storage element (like a resistor-capacitor circuit or temperature system). Its standard transfer function is:

$$G(s) = \frac{K}{\tau s + 1}$$

- K = system gain
- τ = time constant
- Characteristics:
  - Exhibits exponential rise or decay
  - No oscillations
  - Settling time depends on τ

When a PID controller is applied, it improves response speed, reduces steady-state error, and can provide faster settling without overshoot (depending on tuning).

3. Second-Order System

A second-order system has two energy storage elements (e.g., mass-spring-damper system). Its standard transfer function is:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- ω_n = natural frequency
- ζ = damping ratio
- Characteristics:
  - May exhibit overshoot and oscillations if underdamped ($\zeta < 1$)
  - Settling time and overshoot are influenced by ζ and ω_n

PID control can reduce overshoot, improve settling time, and eliminate steady-state error.

4. PID Controller Tuning

The PID controller parameters can be tuned using:

1. **Manual tuning** – trial-and-error method
2. **Ziegler-Nichols method** – based on ultimate gain and oscillation period
3. **MATLAB tools** – such as pidtune for automatic tuning

Effects of PID terms:

- **Proportional (Kp):** Increases response speed, reduces rise time, but may cause overshoot
- **Integral (Ki):** Eliminates steady-state error, may increase overshoot
- **Derivative (Kd):** Reduces overshoot and improves stability, acts as a damping factor

5. Response Study

In MATLAB/Simulink, the following responses are studied:

- **Step Response:** How the system reacts to a sudden change in input
- **Impulse Response:** Reaction to a short pulse
- **Effect of tuning:** Observing changes in rise time, overshoot, settling time, and steady-state error

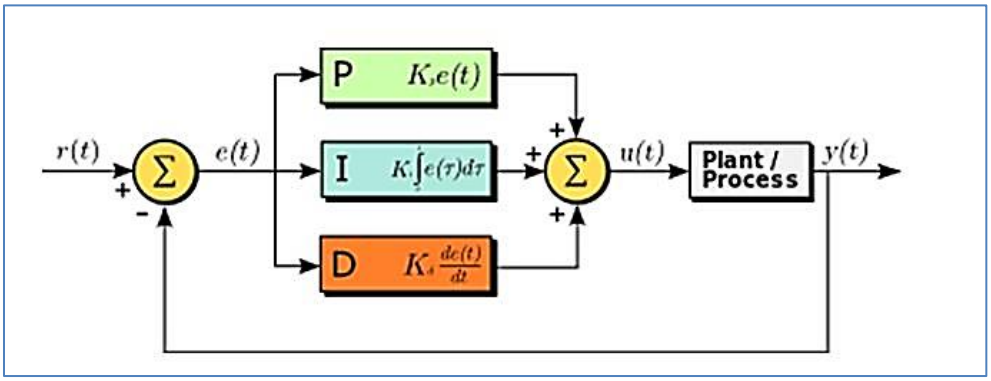
6. PID Controller Implementation in MATLAB/Simulink→

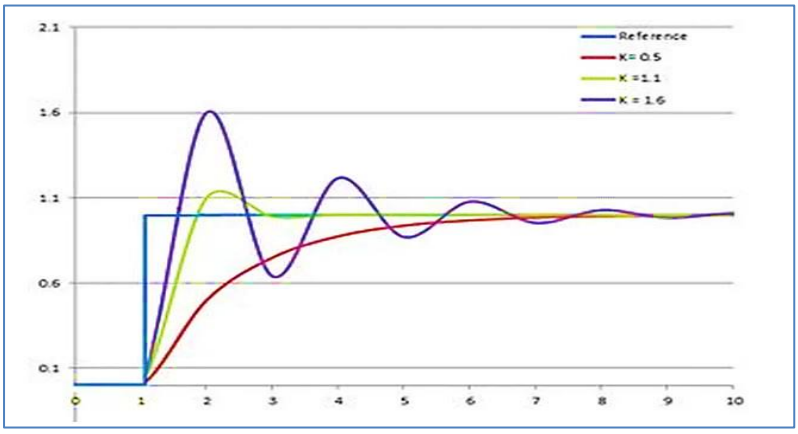
1 Open MATLAB/Simulink

1. Launch MATLAB.
2. Open Simulink by typing `simulink` in the MATLAB command window.
3. Create a **New Model**.

2 Create First-Order System

1. In Simulink Library Browser, drag the following blocks:
   - **Step** (under Sources)
   - **Transfer Function** (under Continuous)
   - **PID Controller** (under Continuous → PID Controller)
   - **Scope** (under Sinks)
   - **Sum** (optional, for feedback loop if using manual connection)
2. Configure **Transfer Function** for first-order system:
   - Numerator: `[K]` (example: `[5]`)
   - Denominator: `[τ 1]` (example: `[2 1]`)
3. Connect Step → PID Controller → Transfer Function → Scope
4. Set **PID Controller parameters**:
   - `Kp = 1`
   - `Ki = 1`
   - `Kd = 0` (start with simple tuning)





RESULT:

Put the screenshot of the result of the experiment performed.(Each result page should contain Name, Roll No & Batch Name)

CONCLUSION:

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